OURANIA RAHMAN

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EDUCATION

Fairleigh Dickinson University, Teaneck, NJ B.S. Dec 2027 | M.S. Dec 2028 (expected)
Honors B.S./M.S. Computer Science (4+1), Cybersecurity & Information Assurance • Minors: Mathematics, Biology, Psychology
GPA 3.95 • Dean's List (consecutive) • Honors Program • Global Scholars Program • CAE-CD Student Affiliate (NSA/DHS)
Honors Thesis: AI Security in SaaS & Cloud Systems (in progress; advisor Dr. Neelu Sinha)
Coursework: Cryptography, Information Security, Firewalls & IDS, Linux Administration, Database Systems, Probability & Statistics, Linear Algebra
Certifications: eCornell Machine Learning Foundations and Agentic AI (2026) • CompTIA Security+ (expected Jan 2027)

Bergen Community College, Paramus, NJ 2022 – 2024
A.S. Biology • **GPA 4.0** • Dean's List every semester • STEM Student Scholars (3SP)

TECHNICAL SKILLS

Languages & ML: Python, Java, JavaScript, C, C++, C#, SQL, Bash, scikit-learn, TensorFlow, pandas, NumPy
Security: Post-quantum cryptography (ML-KEM, ML-DSA), NIST RMF, OWASP Top 10, threat modeling, IAM, SIEM (Splunk, Elastic), firewalls/IDS
AI & Systems: LLM workflows and evaluation (RAG), OpenAI Agents SDK, MCP, static analysis, Akash GPU compute, Git, Linux, M365, TCP/IP, VPN

RESEARCH

DREU Research Fellow (CRA-WP), Emerging Software Technologies Lab, UMBC | Remote Jun – Aug 2026
- Built a multi-stage pre-filter extending the AQuA framework's detection pillar for LLM-based identification of quantum-vulnerable code, raising **precision from 71.98% to 82.55%** and **F1 from 83.71% to 90.44%** while holding **recall at 100%** with zero true positives lost
- Root-caused the precision gap across 15 cryptographic classes (RSA, ECDSA, ECDH, 3DES, SHA-1) and shipped three Python filter modules eliminating 53 false positives, evaluated with per-class precision reporting
- Wrote a 30+ reference literature review; first-author workshop paper in preparation for Q-SE (ICSE 2027)

Secure-by-Design Tool-Use Agent | Research in Computer Science (Dr. Neelu Sinha), FDU Jan 2026
- Built a sandboxed Python AI agent; exposed path traversal and privilege escalation risks, then closed them with allowlisting and audit logging

ML Signal Classification, Optical (Laser) SETI Project, FDU | ML pipeline improving laser-signal detection accuracy Spring 2026 – Present

Undergraduate Researcher, 3SP, Bergen Community College | Lab research on antimicrobial and anticancer properties of fungi Summer 2024

EXPERIENCE

Information Assurance Intern, Department of War (DoW), Fort Meade, MD Summer 2026
- Selected from 1,000+ applicants for a competitive federal cybersecurity scholarship and internship in information assurance
- Offered a Summer 2027 placement in cloud and security architecture with a federal AI office

Break Through Tech AI Studio Fellow, Cornell Tech May 2026 – Present
- Selected from 4,500+ applicants; building an industry-sponsored LLM solution evaluated by F1, ROC-AUC, and subgroup analysis

IT Service Desk Agent, Metro IT Department, FDU, Teaneck, NJ Jan 2026 – Present
- Resolve 400+ tickets monthly and triage 20+ security incidents weekly across IAM, M365, and endpoints; cut resolution time 15%
- Perform security intake for phishing, suspicious sign-ins, and compromised accounts under IR playbooks; enforce least-privilege IAM

Researcher / Teaching Assistant, K-12 Cybersecurity Education Project, FDU Sep 2026 – Present

PROJECTS

Team Lead, Control Systems & Autonomy, NASA RASC-AL Spring 2026 – Present
- Lead a 5-member sub-team in a 23-person engineering group as sole CS contributor, architecting system design and autonomous control logic
- Built rover software from scratch: navigation, telemetry, control, and fault-handling modules delivered ahead of the mid-project design review

Founder, Athena: AI-Powered Identity Risk Monitoring (FDUPitch 2026, 3rd Place) Spring 2026 – Present
- SaaS platform flagging dormant admins, excess privileges, missing MFA, and vendor risk via M365 and Google Workspace APIs
- Sized a ~13,800-firm market from SEC and Verizon DBIR data; defined tiered pricing; leading an expanded team into FDUPitch 2027

PRESENTATIONS

IEEE Quantum Week (QCE26), Toronto, Sept 2026: lightning talk • **Great Minds in STEM (GMiS),** Albuquerque, Sept 2026: PQC research
CAE in Cybersecurity Community Symposium, Pittsburgh, Apr 2026: invited talk on AI and cybersecurity in healthcare
WiCyS 2026, National Harbor, MD, Mar 2026: national poster competition • **IEEE Summit on Applied AI,** Dec 2025 & May 2026: two papers

LEADERSHIP & SERVICE

Founder & President, FDU Baja SAE Chapter (Aug 2025 – Present): lead a multidisciplinary team; manage budget and sponsorships
E-Board Member, IEEE CIS/Systems Council (2026 – Present) • **Student Advisory Board,** Olsen College of Engineering & Science (2026 – Present)
Student Ambassador, Akash Network (Jul 2026 – Present) • **JerseySTEM Instructor** (Oct 2026 – Present): STEM for middle school girls
Volunteer, Bright Side Manor (Jun 2026 – Present) • **Student Alumni Trustee,** BCC (2025) • **Volunteer,** DEF CON 34 Payment Village (Aug 2026)
Peer Mentor, FDU and BCC • **Mathematics Tutor,** FDU (Fall 2025)

HONORS & MEMBERSHIPS

Honors: AnitaB.org Kamala Scholar • NSF S-STEM • ISACA Foundation • HSF • NCA See Yourself in Cyber • WISP • WiCyS Conference Scholar • Phi Theta Kappa • Psi Chi • Tri-Beta
Competitions: 3rd Place, FDUPitch 2026 • Top 3 Regional, IEEEXtreme 19.0 • Top 10, Kean/WiCyS Hackathon
Memberships: CAHSI • IEEE • ACM • SWE • WiCyS • NCSA • NCAA Division I Athlete, Women's Fencing (through Spring 2025)